
DSC 250 Final Report: Metacognition Automatic Detection and Measurement in Student Texts Using Large Language Models

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1 Introduction

Metacognition, commonly referred to as "thinking about thinking" [1], is the awareness of one's own thinking [2]. Metacognitive behaviors involve three processes: planning, self-monitoring, and self-evaluating one's understanding and thought process. Although metacognitive skills have been shown to improve learning approaches and dispel faulty conclusions in educational contexts [3], a key limitation in metacognition research is the scarcity of methods for automatically detecting and studying these behaviors [4]. To address this gap, we successfully developed a LLM-based detector to automatically detect and classify metacognitive behaviors in students' writing, without experimental manipulation or prompting. This will help mitigate the biases that may occur under experimental conditions, such as the observer-expectancy effect, and provide opportunities to analyze the relationship between students' metacognitive behaviors and their academic performance. Our work will also help instructors predict students' academic performance and adjust teaching to better support individual students' need.

2 Related Work

Broadly, metacognition means "thinking about thinking" [1]. Metacognitive behaviors refer to higher-order thinking processes that involve awareness, control, and regulation of one's cognitive processes in learning or problem-solving tasks. The commonly accepted metacognitive behaviors include planning, self-monitoring, and self-evaluating [5, 6].

- **Planning:** refers to the appropriate selection of strategies and the correct allocation of resources that affect task performance.
- **Self-monitoring:** refers to one's awareness of comprehension and task performance.
- **Self-evaluating:** refers to appraising the final product of a task and the efficiency at which the task was performed. This can include re-evaluating strategies that were used.

Previous studies have identified the relationship between practicing metacognition and improved student outcomes, including higher achievement, better learning strategies, and improved programming skills [7, 8, 9]. However, metacognitive strategies are rarely taught or even mentioned explicitly

in college courses [10]. Following foundational research on metacognitive skills, more empirical interventions of metacognition are also being studied.

In our prior work, we developed a tool, named CompassX, to promote all three phases of metacognition [11]. Our records also indicate that feature usage on CompassX and learning outcomes are correlated. Such correlation imply the potential of CompassX to serve as a measurement tool that reflects students’ level of metacognition in the future. To achieve this, we plan to study how to further detect and measure metacognition data more accurately.

Automatic detection method is critical for enabling large-scale metacognition analysis in learning context. However, there are very few studies that focus on metacognition automatic detection and measurement before [4]. Traditional methods for metacognition detection include self-report, observation [12, 13]. But they can not be extended to large-scale and generalizable analysis. Many prior works used natural language processing (NLP) methods to detect related cognitive and affective constructs in student-generated texts [14, 15]. In 2019, Huang et al. [4] proposed a tool for identifying metacognitive language in students’ discussion forum posts. They developed a simple pattern-matching algorithm based on NLP that can locate metacognitive phrases in a body of text [4]. Although Huang et al. [4]’s method allows large-scale analysis of metacognition in online-learning contexts and decreases the time cost, it may face the challenge of insufficient generalization ability. And the bias in their dataset may lead to some inherent limitations, such as potential errors in assigning metacognition. So our project plans to explore the use of LLMs to detect and classify Metacognition in student-produced text within learning contexts, and propose an LLM-based method to solve these problems and improve generalization ability and model interpretability.

3 Methods

3.1 Data Collection

We access a dataset of student transaction logs and think-aloud recordings from three intelligent tutoring systems collected by Zhang et al. This dataset records students’ actions during learning, such as problem-solving attempts and hint requests. The think-aloud transcripts were annotated using a four-stage Self-Regulated Learning (SRL) model (Processing Information, Planning, Enacting, and Realizing Errors) to capture critical cognitive behaviors at each stage of problem-solving. This dataset provides comprehensive information on students’ behaviors at different stages of problem-solving, allowing us to measure metacognition at various phases of students’ learning. The authors annotated 955 items in the dataset, which includes transaction IDs, platform details, anonymized student IDs, students’ utterances, and binary labels (0/1) indicating whether the utterances involve process, planning, action, or error-related thinking. A summary of the dataset is provided in Table 1.

Table 1: Summary of Original Dataset

SRL Category	Behavior	Example	# of Items
Processing Information	- Assemble information - Comprehend information	"Our result should have three significant features. Figures. Avogadro number is 6.02 E23." "2 atoms of H2O."	179
Planning	Identify goals and form plans	"Our goal of the result is hydrogen atoms." "The goal of the result is the number of hydrogen atoms, right?"	125
Enacting	- Verbalize previous action - Announce the next action	"Two molecules of this. How many atoms in a... How many atoms in a minimum molecule of M mole? 61023 divided by 2. 3.0115."	237
Realizing Errors	Realize something is wrong	"It’s incorrect. What’s happened? It is the thousand in the wrong spot. 32 grams per mole. No, the..."	98

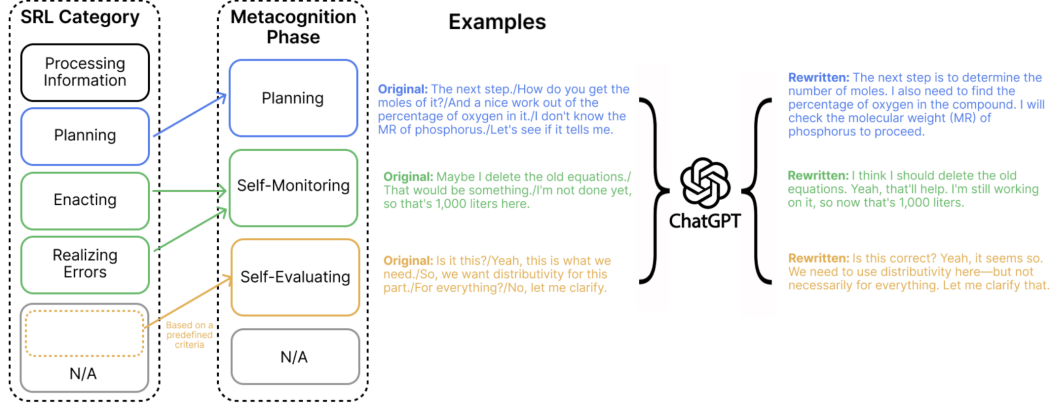


Figure 1: Data Processing Steps and Examples

3.2 Data Processing

To refine the labels of dataset to align with metacognition phases. The *Processing Information* category is excluded from our analysis because it is difficult to directly fill into any of the existing metacognition phases (planning, self-monitoring, self-evaluating). The *Planning* category from the original SRL dataset was retained as the Planning phase in our metacognition framework. Additionally, the *Enacting* and *Realizing Errors* categories were merged to form the Self-Monitoring phase, as both involve students actively engaging with the problem-solving process and identifying mistakes.

Since the original SRL dataset does not include an *Evaluating* category, we manually identified items that belong to the Self-Evaluating phase based on a set of predefined criteria. These predefined criteria were inspired by the annotation guide from Huang et al., which provided valuable insights into identifying reflective and evaluative statements. To be more specific, an utterance is considered self-evaluative if it is expressed in the first person and summarizes task outcomes (e.g., success or failure), analyzes causal factors (e.g., attribution of results), or suggests improvements for future attempts. Additionally, self-evaluative statements typically contain past-oriented language, such as “finally found,” “because,” or similar reflective expressions. To facilitate identification, we considered the presence of keywords commonly associated with self-evaluation, including: *evaluate*, *result*, *outcome*, *concluded*, *cause*, *the main reason is*, *next time*, *learned that*, etc. Using these criteria, we systematically reviewed and labeled utterances that exhibited self-evaluative thinking. When an utterance does not fit into any of the above categories, we classify it as N/A, indicating the absence of metacognitive behavior. As a result, we obtained a dataset with four categories for further use in metacognition automatic detection and classification.

Finally, we found that the verbal nature of text makes it fragmented and informal, and still contains some translation errors. These differences may affect its alignment with the actual text in CompassX. To address this, we used ChatGPT to convert the verbal text into a clearer, more structured format while keeping the student’s natural tone. The data processing steps and examples of text reformatting are shown in Figure 1.

3.3 Prompt Engineering

We designed and refined different prompt formats and examples to direct the LLM in accurately detecting and classifying metacognitive phases. The first stage is prompt pre-training, where we introduce the LLM to metacognition by providing explanations and partial datasets corresponding to different phases. The goal is to help the LLM to learn and summarize the characteristics and criteria of each phase, which will serve as the foundation for the following classification. In the second stage, our baseline prompt will include a brief introduction of metacognition, the classification criteria generated in the first stage, a task description (Task), user input text (Text), and a structured output format, as shown in Figure 2. For the next step, we will explore different prompt word modifications to further improve classification accuracy.

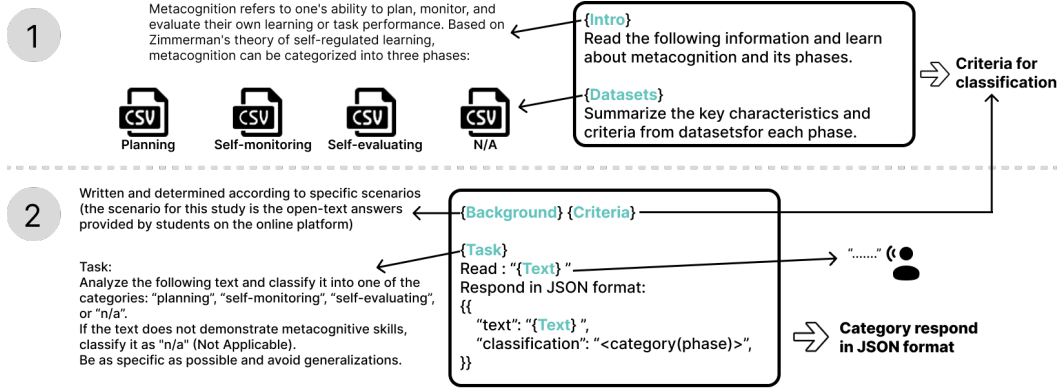


Figure 2: Prompt Design

3.4 Classification

For classification task, we selected 80 samples each from the monitor, planning, and evaluating datasets. Then, we conducted batch experiments with 5 different prompt engineering techniques, and compare with the baseline which only used a simple prompt without examples and details. Conceptually, as we include more diverse and relevant examples in the prompt, the model's accuracy and performance tend to improve.

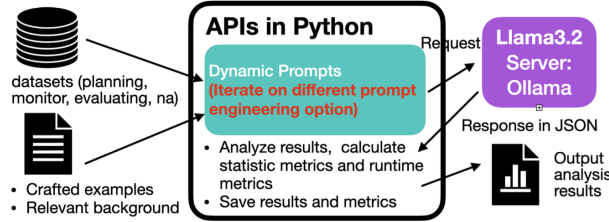


Figure 3: Pipeline for prompt-based classification tasks

The overall classification pipeline is shown in Figure 3. We crafted the prompts by incorporating both the datasets and detailed examples, sending them to the local LLaMA 3.2 server, and then parsed the returned results. The experiment includes five fundamental modes: "basic", "1_ex", "5_ex", "10_ex", and "chain of thought". These modes correspond to different prompt engineering strategies listed below:

- "Basic": This serves as an enhancement of the baseline method by concatenating an example set.
- "1_ex", "5_ex", "10_ex": These modes align with the previously introduced prompt engineering strategies. (using one, five, and ten examples respectively)
- "Chain of thought": This mode requires the model to think step-by-step and include a "chain of thought" field in the returned JSON file.

Additionally, we also implemented the similar pipeline for binary classification between *monitor* and *na*. The key distinction lies in our use of specialized prompts that incorporate detailed descriptions to address specific data limitations. This approach allows us to mitigate issues arising from varying dataset quality.

3.5 Evaluation

The classification task aimed to categorize student responses into four labels: Planning, Monitor, Evaluating, and NA (non-metacognitive text). Different prompt engineering strategies were employed, including basic prompts without examples given, 1-shot, 5-shot and 10-shot examples. To evaluate the performance of the classification model, we employed the following metrics:

- Precision: Measures the proportion of correctly classified instances within a given category relative to all instances predicted as that category.
- Recall: Assesses the proportion of true instances of a category that were correctly identified by the model.
- F1-score: The harmonic mean of Precision and Recall, providing a balanced measure of model performance.
- Accuracy: Represents the proportion of correctly classified instances across all categories.

To further analyze the model’s classification behavior, we utilized confusion matrices to visually illustrate its performance across different categories and identify misclassification patterns.

4 Results

The classification experiments in this study evaluated various prompt engineering techniques for classifying student responses regarding their metacognition phrase.

As shown in Figure 4, the confusion matrices illustrate the model performance under different prompting strategies for the classification of all four categories. The detailed results are also summarized in Table 2.

Table 2: Performance metrics for different modes

Mode	Label	Precision	Recall	F1-Score	Accuracy
basic	planning	0.4872	0.5000	0.4935	0.8395
	monitor	0.4741	0.8208	0.6011	0.6996
	evaluating	0.5714	0.2069	0.3038	0.7737
	na	0.8955	0.7500	0.8163	0.8888
1_ex	planning	0.5000	0.5641	0.5301	0.8402
	monitor	0.4766	0.7612	0.5862	0.7049
	evaluating	0.4762	0.1724	0.2532	0.7582
	na	0.8888	0.8000	0.8421	0.9016
5_ex	planning	0.6053	0.5897	0.5974	0.8730
	monitor	0.4828	0.8358	0.6120	0.7090
	evaluating	0.5833	0.2414	0.3415	0.7787
	na	0.9091	0.7500	0.8219	0.8934
10_ex	planning	0.5676	0.5385	0.5526	0.8607
	monitor	0.4545	0.8209	0.5851	0.6803
	evaluating	0.7368	0.2414	0.3636	0.7992
	na	0.9403	0.7875	0.8571	0.9139
COT	planning	0.5882	0.5128	0.5479	0.8642
	monitor	0.4754	0.8406	0.6073	0.6914
	evaluating	0.5500	0.1930	0.2857	0.7737
	na	0.9552	0.8205	0.8828	0.9300

The classification accuracy of the planning label is the highest, reaching 0.93 under the chain-of-thought prompting strategy. The highest F1-Score is 0.8828. For the evaluating label, the highest accuracy is 0.7992 while the highest F1-Score is only 0.3636, both achieved under the 10-shot examples prompting strategy. The planning and monitor category achieved highest performance under 5 shot strategy, with the highest F1-Score of 0.5974 and 0.6120 respectively, and highest accuracy of 0.8730 and 0.7090 respectively. Providing a few examples in the prompts helps the model classify phrases more accurately. However, more examples do not necessarily lead to better performance. Using 5–10 examples may yield the best results, as an excessive number of examples can lengthen the context, potentially distracting the model from key information. The predictions for evaluating are the least accurate, indicating that LLaMa has the weakest understanding of this behavior.

Confusion Matrices for Multi-Classification Task Under Different Tactics

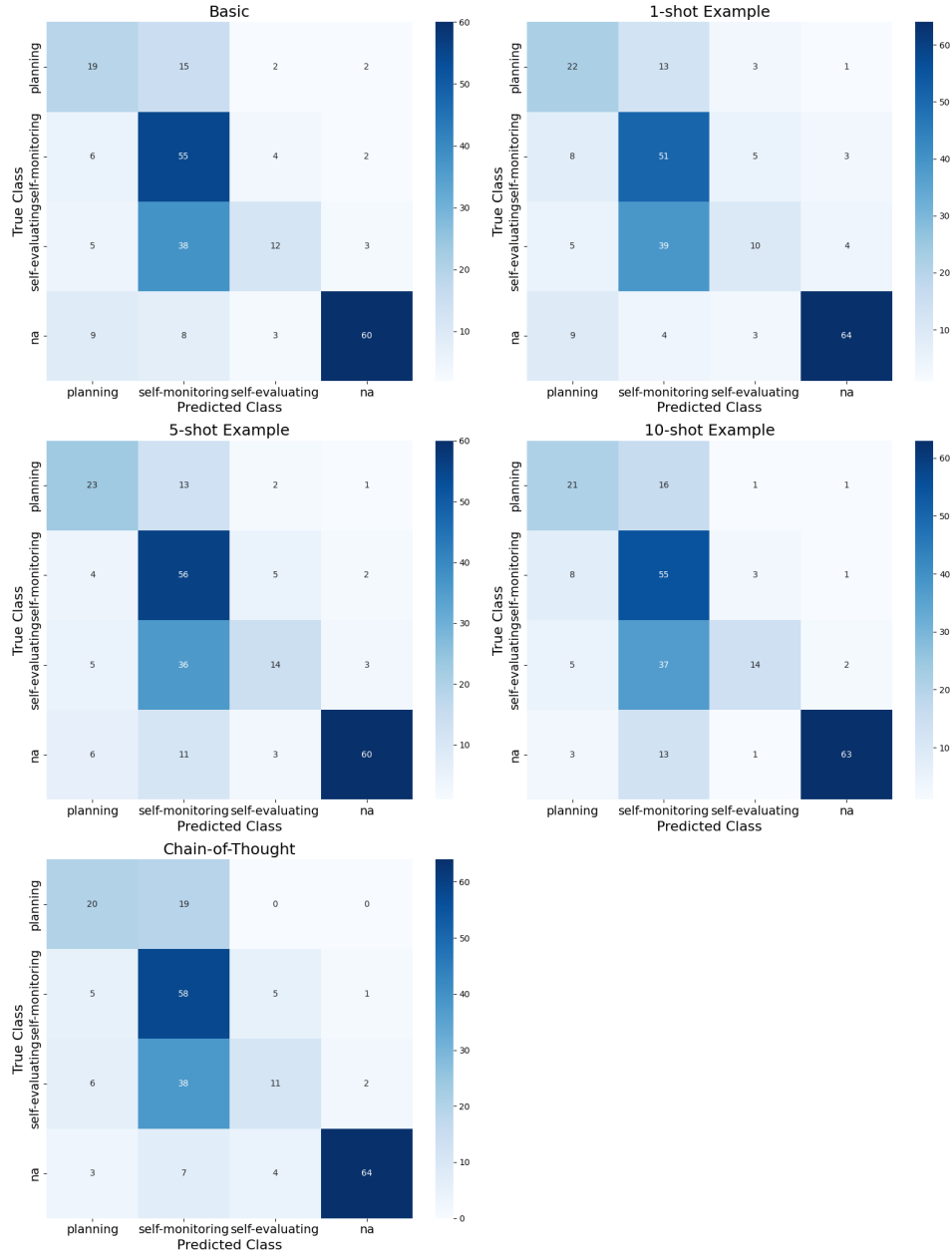


Figure 4: Confusion matrices for multi-classification task under different tactics

To validate the effectiveness of verbal-to-written conversion and the prompt engineering, we compared LLaMa’s classification performance on the original texts with simple prompts and converted texts with experimental prompts. We evaluated the best-performing 5-shot examples in both cases, with the confusion matrices shown in the Figure 5.

The results indicate that the model performs poorly across all four categories on the original methods, with a strong tendency to classify most samples as NA. This may be due to the conversational nature of the original text, which makes its metacognition features less distinct. In contrast, verbal-to-written conversion significantly improves classification accuracy for each category, and with the assistance of improved prompts.

Confusion Matrix Heatmap for Mode: 5_ex

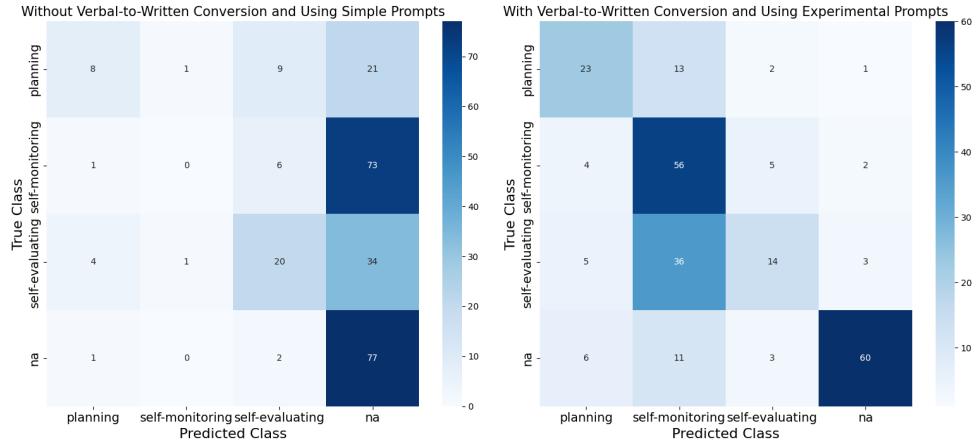


Figure 5: Confusion matrices for multi-classification task under different tactics

5 Limitation & Future Work

5.1 Discussion & Limitation

Our work also presents limitations that should be addressed in future research.

One limitation is the quality of the dataset used in this study, as it consists of students’ think-aloud transcripts. The verbal nature of the dataset makes it fragmented and includes informal language, as well as grammar errors. Additionally, there is an insufficient and unbalanced number of labeled data items, which could potentially lead to biased results and limited generalizability. Furthermore, the domain of the dataset, Stoichiometry and chemistry, is not directly related to the field of computing. Therefore, the model might not preserve transferability to computing-related fields.

To address these limitations, we plan to collect our own dataset in Spring quarter using CompassX, a tool developed by our team to collect computing students’ metacognition data. It can collect students’ text artifacts from approximately 300-400 students and is employed in an introductory-level data science course focusing on computing algorithms.

Additionally, the performance of our model based on the current prompt is not yet optimal, as the generated responses may not always be sufficiently accurate or relevant, which can also affect the effectiveness of the model. To address this, we plan to convert labeled examples into vector representations and employ similarity matching techniques. This approach will help us retrieve the most relevant examples for use as prompts. We expect this method will further improve the quality of the model’s output.

5.2 Future Direction

After obtaining a higher quality dataset using CompassX, our first step will be to conduct a comparative analysis between the LLM-based detector and traditional machine learning (ML) methods. We plan to evaluate the performance of both approaches in terms of accuracy, speed, and generalizability, while identifying the strengths and weaknesses of each method in handling challenges within the dataset, such as fragmented sentences and non-standard grammar.

Upon successfully developing the detection and classification algorithm, the next goal is to design a study to understand the relationship between university students’ metacognition and their performance. There are several directions we can pursue:

1. **Extend our model’s ability to predict academic performance based on students’ mastery of metacognitive skills.** Prior studies suggest that high-achieving students tend to be more aware of cognitive rules and evoke metacognitive knowledge in their reflection [16]. Additionally, their metacognitive knowledge appears more structured and hierarchically

organized [16]. Therefore, future research can focus on understanding the metacognition-performance relationship to predict academic outcomes.

2. Investigate the self-confidence factor and its relationship with metacognitive measures.

Additionally, examine how students' confidence varies during each phase of the metacognition process. Prior work indicates that the degree of association between accuracy and confidence can be used as a quantitative measure of metacognition [17]. Several model-based approaches, such as the Meta-d' approach and the Stochastic Detection and Retrieval Model (SDRM), have been employed to measure metacognition [18]. Given that confidence can also be assessed using self-report questionnaires, we can integrate these assessments to better understand the relationship between students' self-reported confidence and their actual metacognitive behaviors.

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